

BOLUNGAVIK, IcelandWMO: **040050**Lat: **66.161N**Lon: **23.254W**Elev: **27**StdP: **101.00**Time Zone: **0.00 (E00)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.4	-7.9	-13.9	1.1	-8.3	-12.3	1.3	-7.0	20.2	-1.2	18.0	-0.4	5.1	240	0.801

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.8	16.0	11.9	14.8	11.2	13.8	10.6	12.6	15.1	11.7	14.1	11.1	13.3	3.6	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
11.1	8.3	13.7	10.3	7.8	12.9	9.6	7.5	12.3	35.7	15.2	33.6	14.2	32.1	13.4	17.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	13.8	12.0	DB	-11.2	18.5	2.0	1.6	-12.6	19.7	-13.8	20.6	-14.9	21.4	-16.4	22.6	(4)
(5)			WB	-12.1	14.5	1.8	1.2	-13.4	15.3	-14.5	16.0	-15.5	16.6	-16.9	17.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	0.0	-0.6	-0.5	1.6	4.6	8.2	10.1	9.7	7.4	3.6	1.3	0.1	(6)	
	DBStd	5.04	3.48	3.71	3.76	3.24	2.82	2.55	2.23	2.21	3.16	3.29	3.31	3.61	(7)	
	HDD10.0	2330	310	297	324	252	168	64	26	32	89	199	262	307	(8)	
	HDD18.3	5294	568	530	583	502	426	303	256	267	327	457	512	565	(9)	
	CDD10.0	78	0	0	0	0	1	11	29	24	12	1	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	4.7	5.4	5.5	5.4	4.7	4.1	3.8	3.3	3.4	4.1	5.3	5.3	5.6	(14)	
(15) Precipitation	PrecAvg	1092	126	120	105	66	49	32	46	62	104	126	127	135	(15)	
	PrecMax	1647	226	331	179	133	112	94	116	108	164	212	208	243	(16)	
	PrecMin	814	57	44	36	26	13	1	14	17	33	44	55	35	(17)	
	PrecStd	188	43	58	36	27	26	21	22	27	37	49	44	46	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	9.2	9.9	11.6	13.7	16.7	18.0	17.3	16.2	13.6	11.9	10.2	(19)	
		MCWB	7.0	6.4	7.0	7.9	9.0	11.7	12.4	13.1	12.4	10.3	9.1	7.2	(20)	
	2%	DB	8.1	7.9	7.9	9.2	11.4	14.8	16.1	15.4	14.6	11.4	9.2	8.3	(21)	
		MCWB	5.7	5.5	5.6	6.3	8.0	10.5	11.9	11.8	11.4	8.4	6.6	5.8	(22)	
	5%	DB	6.4	6.3	6.3	7.6	10.1	13.4	14.8	14.2	13.3	9.7	7.1	6.6	(23)	
		MCWB	4.3	4.2	4.3	5.1	7.2	9.7	11.2	11.2	10.5	7.4	5.2	4.6	(24)	
	10%	DB	5.0	4.7	4.7	6.3	8.9	12.2	13.7	13.1	12.0	8.4	5.8	5.2	(25)	
		MCWB	3.0	2.8	2.8	4.2	6.4	9.1	10.5	10.6	9.5	6.5	4.2	3.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	6.9	7.2	8.2	9.8	12.2	13.7	13.8	13.0	10.6	9.2	7.7	(27)	
		MCDB	9.3	8.4	9.5	11.1	12.9	15.4	16.7	16.2	15.6	13.6	11.7	10.2	(28)	
	2%	WB	5.8	5.6	5.8	6.5	8.3	11.2	12.5	12.5	11.7	8.7	6.7	6.0	(29)	
		MCDB	7.7	7.7	7.8	9.0	10.8	14.0	15.2	14.6	14.1	10.8	8.8	8.1	(30)	
	5%	WB	4.5	4.3	4.5	5.3	7.4	10.2	11.7	11.7	10.7	7.7	5.4	4.8	(31)	
		MCDB	6.3	6.2	6.2	7.4	9.8	12.8	14.0	13.6	13.0	9.5	7.0	6.4	(32)	
	10%	WB	3.2	3.0	2.9	4.3	6.5	9.3	11.0	10.9	9.8	6.6	4.3	3.6	(33)	
		MCDB	4.9	4.5	4.5	6.2	8.7	11.8	13.2	12.7	11.7	8.2	5.7	5.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.2	4.2	4.3	4.5	4.8	4.8	4.5	4.1	3.6	3.9	4.2	(35)	
		MCDBR	5.4	6.1	5.6	6.0	6.4	6.9	7.1	6.2	5.4	5.2	5.9	5.7	(36)	
	5% WB	MCWBR	4.5	5.0	4.6	4.2	4.3	4.2	4.1	3.9	4.0	4.0	5.0	4.6	(37)	
		MCWBR	5.3	6.0	5.6	5.9	5.9	6.1	6.2	5.5	5.2	4.8	5.7	5.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.388	0.291	0.319	0.346	0.347	0.337	0.347	0.344	0.337	0.288	0.366	0.380	(40)	
		taud	2.532	2.143	2.260	2.272	2.394	2.476	2.476	2.502	2.416	2.255	2.648	2.632	(41)	
	Ebn at Noon		191	550	738	812	854	874	852	818	724	555	220	55	(42)	
		Edh at Noon	21	54	79	101	99	94	91	81	69	47	19	9	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.06	0.36	1.26	2.77	4.21	5.07	4.38	3.31	1.77	0.59	0.11	0.01	(44)	
	RadStd		0.00	0.06	0.15	0.26	0.49	0.62	0.42	0.33	0.15	0.06	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.68	+1.52	+1.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		+0.68	+1.52	+1.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page